Digital Communication (Week 2.1)

(i) Random Variables
(ii) Probability Distribution Function
(iii) Cumulative Distribution Function (CDF)
(iv) Expectation, Moments

What is the probability that the height of a random student is less than a particular value?

Let's create a table for it. We are considering 8 samples here.

	4/8 0.5	5/8 0.75	6/8 1.0	8/8 1.6
Heights of Students	≤4	≤5	≤6	≤7
5	X	✓	✓	✓
3.1	✓	✓	✓	✓
3.9	✓	✓	✓	✓
4.1	X	✓	✓	✓
6	X	X	✓	✓
3.5	✓	✓	✓	✓
3.8	✓	✓	✓	✓
5.5	X	X	✓	✓

Observation

(i) Probability that is in range [0,1].
(ii) Cumulative in nature.
(iii) Non-decreasing in nature.
(iv) Saturation point.

Here, we proceed in two approaches -

(a) Values less than 4 is already less than 5, so no comparison needed for them.

(b) Values that are not less than 4, needs to be rechecked.

Here, height is called a random variable.

discrete random variable.

Takes values from a discrete dataset randomly.

This is just an example. We are interested in continuous random variable.

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Say, X is a continuous Random Variable

Variable that takes values from some continuous dataset in a random fashion.

The Probability Distribution Function or Cummulative Distribution Function (CDF) can be defined by

$cdf(x_0) = Pr(X \leq x_0)$, where, cdf is the cummulative distribution function

x_0 is a particular value from the considered sample space

X is the random variable

Pr is probability.

Properties

① $0 \leq cdf(x) \leq 1$, $\forall x$ as it is a probability.

② $cdf(+\infty) = 1$, probability of anything smaller than $+\infty$ is 1, $Pr\{X \leq +\infty\} = 1$.

③ $cdf(-\infty) = 0$, probability of something smaller than $-\infty$ is 0, $Pr\{X \leq -\infty\} = 0$.

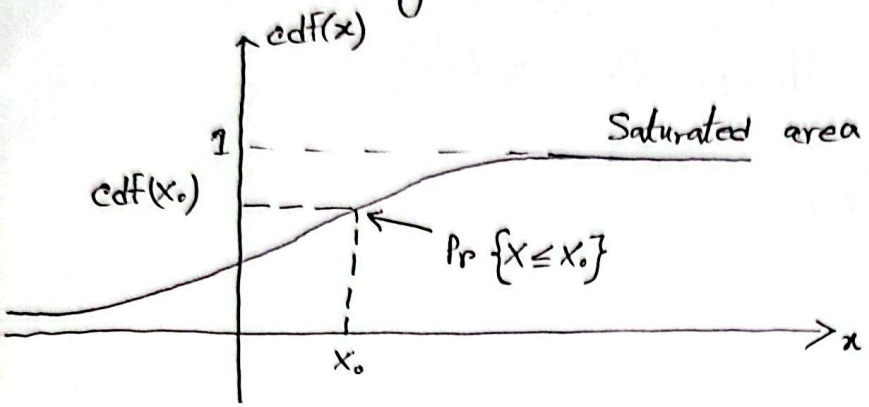
④ $Pr\{a \leq x \leq b\} = cdf(b) - cdf(a)$, as cdf is a cummulative function.

⑤ $Pr\{x > 0\} = 1 - Pr\{x \leq a\}$

$= 1 - cdf(a)$

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Consider a general curve for cdf:



Example: $cdf(x) = \begin{cases} 1 - e^{-2x}, & x \geq 0 \\ 0, & x < 0 \end{cases}$

Determine: ① $Pr\{X \leq 3\}$

② $Pr\{X \in [1,3]\}$

① $Pr\{X \leq 3\} = cdf(3) = 1 - e^{-2.3} = 1 - e^{-6} = 0.9975$

mean, 99.75% of the time the random variable X will take values less than 3.

99.75% of time a numeric property of a system satisfy that it is less than 3.

② $Pr\{X \in [1,3]\} = Pr\{1 \leq x \leq 3\}$

$= cdf(3) - cdf(1)$

$= (1 - e^{-2.3}) - (1 - e^{-2.1})$

$= 1 - e^{-2.3} - 1 + e^{-2.1}$

$= e^{-2} - e^{-6}$

$= 0.1329$

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